

2022

DECEMBER

Su	Mo	Tu	We	Th	Fr	Sa
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

2022 NOVEMBER

Thursday

Week 47 ■ 321-044

17

AAD Mod-5

Problems

Solvable

Unsolvable

Tractable

Intractable

Input size: $\downarrow$ —————→ Exponential fn

Polynomial fn.

P

NP

Verified in Polynomial time

NP Hard:

$Y \propto X$
 $\downarrow$

NP Hard if Y is NP Hard.

⇒ for every Y in NP, it is possible to polynomially reduce it to X .

↓

transformation one instance in Y to an instance in X , taking polynomial time.

if X takes polynomial time

then Y takes polynomial time

IF NP & NP-Hard $\rightarrow$ NP-Complete.

NOVEMBER 2022

18



Friday

Week 47 ■ 322-043

OCTOBER							2022
Su	Mo	Tu	We	Th	Fr	Sa	
30	31						1
2	3	4	5	6	7	8	8
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16	17	18	19	20	21	22	22
23	24	25	26	27	28	29	29

Approximation algorithm

- Feasible, close to optimal.
- Takes polynomial time

Approximation ratio / factor:

→ Maximization: $\frac{C^*}{C}$ $0 < C \leq C^*$

→ Minimization: $\frac{C}{C^*}$ $0 < C^* \leq C$

Bin-packing algorithm

Min. no. of bins = $\text{Ceil} \left(\frac{\text{Total weight}}{\text{Capacity}} \right)$

- Next fit
 - First fit
 - Best fit
 - Worst fit
- } Online (order unknown)
- Best fit decreasing
 - First fit decreasing
- } Offline (order known)

Graph coloring problem: Coloring vertices of a graph such that no two vertices adjacent to each other have the same color.

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11	12	13	14	15	16	17	
18	19	20	21	22	23	24	
25	26	27	28	29	30	31	

2022 NOVEMBER
Saturday
Week 48 ■ 330-035



NP Completeness Proof

Clique Problem (Clique: Complete Subgraph)

Prove given problem is both NP & NP Hard.

NP - Write a polynomial time verificatn. algo.

Let $G = (V, E)$, $V' \subseteq V$



has all the k vertices forming the clique

→ Check all ~~the~~ vertices in V' are in G

→ Check each pair $(u, v) \in V'$ has edge (u, v) in E , i.e., ~~check~~ ^{check} all vertices in V' are connected to each other.

Hence proved, if both tests pass.

⇒ Hence NP.

NP-Hard - Write a polynomial time reduction algo.

Sunday 27

Week 49 ■ 331-034

i.e., transform ~~3SAT~~ CNF-SAT to Clique problem.

Let $\phi = C_1 \wedge C_2 \wedge \dots \wedge C_k$
↳ clauses.

NOVEMBER 2022

28

Monday

Week 49 ■ 332-033

OCTOBER

Su	Mo	Tu	We	Th	Fr	Sa
30	31					
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
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→ Let each clause have two distinct literals x_1, x_2, x_3 .

→ Construct graph G such that ϕ is satisfying iff G has a clique of size k .

→ G construction:

→ Each literal is a vertex in G .

→ Vertex connection:

$$E = \{ \langle a, i \rangle, \langle b, j \rangle \mid i \neq j \text{ and } b \neq \bar{a} \}$$

If G has a clique of size k , then ϕ has a satisfying assignment, i.e., vertices that form the clique satisfies ϕ .
 $\Rightarrow$ Hence NP-Hard.

Vertex ~~Pro~~ Cover Problem: Min. no. of vertices required to cover all the edges in a graph.

NP: Polynomial time verification algorithm

1) count = 0

2) for each vertex $v \in V'$, remove all edges adjacent to v from set E

a) increment count by 1.

3) if count = k & $E = \emptyset$, then correct

4) Else wrong

DECEMBER 2022

Su	Mo	Tu	We	Th	Fr	Sa
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4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

2022 NOVEMBER

Tuesday

Week 49 ■ 333-032

29

$\Rightarrow$ Hence NP.

NP-Hard: Polynomial time reduction algo.

Clique $\rightarrow$ vertex cover.

1) Construct complement of G , i.e., G' .

2) If G' has a vertex cover of size $|V| - k$, then G has a clique of k -size.
 $\Rightarrow$ Hence NP-Hard.

Randomised Algorithms

Las Vegas

$\rightarrow$ O/p always correct & optimal

$\rightarrow$ Running time uncertain

Eg:- Randomised Quick Sort

Algorithm findingA_LV(A, n)

{

repeat

{

Randomly choose one element out of n elements

until ('a' is found)

}

NOVEMBER 2022

30

Wednesday

Week 49 ■ 334-031

OCTOBER						
Su	Mo	Tu	We	Th	Fr	Sa
30	31					
2	3	4	5	6	7	8
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JANUARY 2023						
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15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

2022 DECEMBER

Thursday

Week 49 ■ 335-030



Monte Carlo

- May produce correct o/p with some probability.
- Has fixed no. of steps $\Rightarrow$ Deterministic run time.

~~Randomized~~

Algorithm finding $A_MC(A, n, k)$

```

{
    i = 0;
    repeat
    {
        Randomly select one element out of n elements
        i = i + 1;
    } until (i = k or 'a' is found);
}
    
```

Algorithm rand Quick Sort ($A[], low, high$)

- 1) If $low \geq high$, then EXIT
- 2) While pivot 'x' is not a central Pivot
 - 2.1) Choose uniformly at random an element from $A[low \dots high]$.
Let the randomly picked element be x.
 - 2.2) Count elements in $A[low \dots high]$ that are $\leq x$. Let it be sc.
 - 2.3) Count elements in $A[low \dots high]$ that are $> x$. Let it be gc.
 - 2.4) $n = (high - low + 1)$
if $sc \geq n/4$ & $gc \geq n/4$,
 $\Rightarrow x = \text{central pivot.}$
- 3) Partition $A[low \dots high]$ into two subarrays. The first subarray has elements in $A \leq x$ & second, $> x$.
index of x = pos.
- 4) rand Quick Sort ($A, low, pos-1$)
- 5) rand Quick Sort ($A, pos+1, high$)

Probability = $1/n$

Time complexity = $O(n \log n)$.